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Microbial community structure in a thermophilic aerobic digester used as a sludge pretreatment process for the mesophilic anaerobic digestion and the enhancement of methane production

Hyun Min Jang [a], Sang Kyu Park [b], Jeong Hyub Ha [a,b,*], Jong Moon Park [a,b,c,*]

[a] School of Environmental Science and Engineering, Pohang University of Science and Technology, Hyoja-dong, Nam-gu, Pohang, Kyeongbuk 790-784, Republic of Korea
[b] Department of Chemical Engineering, Pohang University of Science and Technology, Hyoja-dong, Nam-gu, Pohang, Kyeongbuk 790-784, Republic of Korea
[c] Division of Advanced Nuclear Engineering, Pohang University of Science and Technology, Hyoja-dong, Nam-gu, Pohang, Kyeongbuk 790-784, Republic of Korea

HIGHLIGHTS

- A combined process was developed for sludge reduction and methane production.
- Biological TAD pretreatment highly increased soluble organic matters.
- Bacteria species using DGGE were examined in a combined process.
- Methanogen species using a real-time PCR were examined in the MAD.
- Methane enhancement by TAD pretreatment was observed in a combined process.

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ABSTRACT

An effective two-stage sewage sludge digestion process, consisting of thermophilic aerobic digestion (TAD) followed by mesophilic anaerobic digestion (MAD), was developed for efficient sludge reduction and methane production. Using TAD as a biological pretreatment, the total volatile suspended solid reduction (VSSR) and methane production rate (MPR) in the MAD reactor were significantly improved. According to denaturing gradient gel electrophoresis (DGGE) analysis, the results indicated that the dominant bacteria species such as *Ureibacillus thermophiles* and Bacterium thermus in TAD were major routes for enhancing soluble organic matter. TAD pretreatment using a relatively short SRT of 1 day showed highly increased soluble organic products and positively affected an increment of bacteria populations which performed interrelated microbial metabolisms with methanogenic species in the MAD; consequently, a quantitative real-time PCR indicated greatly increased *Methanosarcinales* (acetate-utilizing methanogens) in the MAD, resulting in enhanced methane production.

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1. Introduction

The generation of waste activated sludge (WAS) continues to increase rapidly, and so more efficient methods are required for managing such organic materials. In Korea, the number of wastewater treatment plants (WWTP) has increased, and 8295 tons (dry weight) of WAS is generated daily (MOE, 2009). Recently, about 6241 tons (75% of total generation) of WAS per day were discharged or removed by ocean dumping, landfill, and incineration (MOE, 2009). However, these methods of WAS management are increasingly restricted by stringent regulation, rapidly shrinking landfill space, and public opposition. Thus, it is apparent that these methods will not be viable in the near future.

Anaerobic digestion (AD) has been widely used as an efficient means of treating such wastes, since it converts organic waste into biogas in the form of methane, which is a renewable energy source (Speece, 1983). Despite these advantages of AD, some limitations are inevitable; the rather slow digestion rate, long duration of solid retention time (SRT) (20–40 days), and the susceptibility of the process to variations in operating conditions (Chen et al., 2008). Thus, many researchers have focused on increasing digestion rate by promoting hydrolysis, which is generally regarded as the rate-limiting step of AD. Various physical, chemical, thermo-chemical, ultrasonic and freezing and thawing pretreatments have been applied to improve AD efficiency (Carrere et al., 2010). However,

* Corresponding authors at: School of Environmental Science and Engineering, Pohang University of Science and Technology, Hyoja-dong, Nam-gu, Pohang, Kyeongbuk 790-784, Republic of Korea. Tel.: +82 54 279 8315; fax: +82 54 279 8659 (J.H. Ha), tel.: +82 54 279 2275; fax: +82 54 279 8659 (J.M. Park).
E-mail addresses: jeonghha@postech.ac.kr (J.H. Ha), jmpark@postech.ac.kr, jmpk777@yahoo.com (J.M. Park).

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these conventional pretreatment methods are not economical, and especially chemical and freezing and thawing pretreatment methods could reduce methane yield owing to increased salinity and retaining of lipid or protein in the solid phase (Liu et al., 2008).

Autothermal thermophilic aerobic digestion (ATAD) may provide an efficient and environmentally friendly biological pretreatment that can be applied to municipal solid waste (MSW) (Kelly et al., 1993). The ATAD treatment process has advantages such as enteric pathogen inactivation, short hydraulic retention time (HRT), low operating costs and self-heating during microbial metabolism. In addition, it achieves relatively fast degradation of volatile suspended solids (VSS), effectively disrupts the microbial cells of sludge and significantly increases the soluble chemical oxygen demand (SCOD) concentration such as carbohydrate, protein, lipid and VFAs, since thermophilic microbes release proteases that are highly active in sludge degradation, which was the major known lytic enzyme during the hydrolysis of sludge (Yan et al., 2008). In this respect, the interest in ATAD process as sludge pretreatment and improved digestion processes has increased recently.

As another recent technological improvement for the sludge digestion processes, combined processes have been used to achieve high solid reduction. Some researchers reported that thermophilic aerobic digestion (TAD) process prior to mesophilic anaerobic digestion (MAD) showed greater reduction in volatile solids (VS) and pathogens compared to a single MAD process (Hasegawa et al., 2000; Pagilla et al., 1996). Also, two-stage anaerobic thermophilic and anaerobic mesophilic treatment methods are known to provide both efficient VS reduction and gas production (Han et al., 1997). Furthermore, it was suggested that the combined process addresses the notion that certain proportions of sewage sludge can be degraded only under anaerobic or aerobic conditions (Novak et al., 2003). In this respect, the combined anaerobic and aerobic digestion process can promote additional degradation of organic matter, leading to greater reduction in solids compared to a single AD process.

Although the combined process provides a useful alternative method for sludge reduction and methane production, research on the combined process is in its infancy. Thus, the objective of the present study was to investigate the feasibility of a combined TAD–MAD process for the treatment of sludge; to assess the potential advantages (i.e., greater VSS reduction, VSSR and increased methane production rate, MPR); and to ascertain the optimum SRT of TAD within a combined system. In addition, there are no previous reports on the microbial community and population dynamics in continuous combined TAD–MAD process under different operating conditions. Thus, this study also aimed to elucidate the qualitative and quantitative microbial community structures and population changes in the combined TAD–MAD process using denaturing gradient gel electrophoresis (DGGE) and real-time PCR analysis.

2. Methods

2.1. Feed-sludge preparation

The feed sludge for this study was a mixture of primary and secondary sludge samples, collected from the municipal WWTP in Daegu, Korea. The plant's activated sludge process treats 520,000 m^3/d of domestic wastewater. Prior to mixing, all material was filtered by 1.0 mm sieve to remove inert matter, and then the primary and secondary sludge samples were mixed thoroughly (proportions 3:7 v/v). The mixed sludge was transferred to 3-L bottles and stored at −25 °C until use. The physical/chemical characteristics of the feed sludge used in this study are presented in Table 1.

2.2. Reactor setup

The design of the combined process is shown in Fig. 1. It consists of a TAD process (R_1) followed by MAD process (R_2); and a single MAD process (R_3) acting as a control. R_1 serves as a biological pretreatment process, and R_2 as sludge digestion and methane production. R_1 was seeded with sludge from a successfully operated ATAD pilot plant in Daejeon, Korea, while R_2 and R_3 were seeded with mesophilic anaerobic sludge from a treatment plant in Daegu, Korea, respectively. Prior to continuous operation, all reactors were operated in batch mode for two weeks. The reactors were fed four times a day, using a peristaltic pump (Cole-Parmer®) controlled by a timer and relay. During the experimental period, feeding and discharge were conducted simultaneously. R_2 and R_3 were operated at SRT of 40 days in phases I, II and III, while R_1 was operated under different SRTs (4, 2 and 1 day) in order to investigate the effects of TAD pretreatment on anaerobic digestion performance. Meanwhile, in phase IV, R_2 was operated at SRT of 39 days, and R_1 was operated at SRT of 1 day in order to make the total control volume the same as R_3. All reactors were operated as a CSTR with complete mixing, so that HRT and SRT in the system were equal. More detailed operating conditions are described in Table 2.

2.3. Physical and chemical analytical methods

Standard Methods (APHA-AWWA-WEF, 1998) were used to determine total suspended solids (TSS), VSS, total COD (TCOD), total nitrogen (TN), alkalinity and total phosphorus (TP). After centrifugation for 30 min at 5000 rpm, supernatant was filtered by 0.45-μm syringe filter (Whatman, USA) to measure SCOD, ammonia (NH_4^+–N), soluble TN (STN), and soluble TP (STP). The carbohydrate and protein concentrations were measured using the phenol–sulfuric acid method (DuBois et al., 1956) and Lowry–Folin method (Lowry et al., 1951), respectively. The concentrations of nitrite (NO_2^-), nitrate (NO_3^-), and orthophosphate (PO_4^{3-}–P) were determined by ion chromatography (ICS-100, DIONEX Co., USA). The pH and oxidation/reduction potential (ORP) in each reactor were con-

Table 1
Characteristics of feed sludge.

Parameter	Value (average ± S.D.)
pH	6.61 ± 0.06
Alkalinity (g $CaCO_3$/L)	1.72 ± 0.06
TSS (g/L)	51.51 ± 5.22
VSS (g/L)	35.56 ± 2.47
TCOD (g/L)	62.56 ± 7.96
SCOD (g/L)	7.96 ± 0.8
Nitrogen	
TN (g/L)	4.51 ± 0.77
STN (g/L)	0.96 ± 0.12
NH_4^+–N (g/L)	0.90 ± 0.09
NO_2^- (g/L)	–
NO_3^- (g/L)	–
Phosphorus	
TP (g/L)	1.81 ± 0.12
STP (g/L)	0.43 ± 0.02
PO_4^{3-} (g/L)	0.42 ± 0.01
Total VFAs (g COD/L)	1.54 ± 0.16
Acetic acid (g COD/L)	1.54 ± 0.16
Butyric acid (g COD/L)	–
Propionic acid (g COD/L)	–
Isobutyric acid (g COD/L)	–
Isovaleric acid (g COD/L)	–
Protein (g COD/L)	1.81 ± 0.29
Carbohydrate (g COD/L)	0.63 ± 0.13

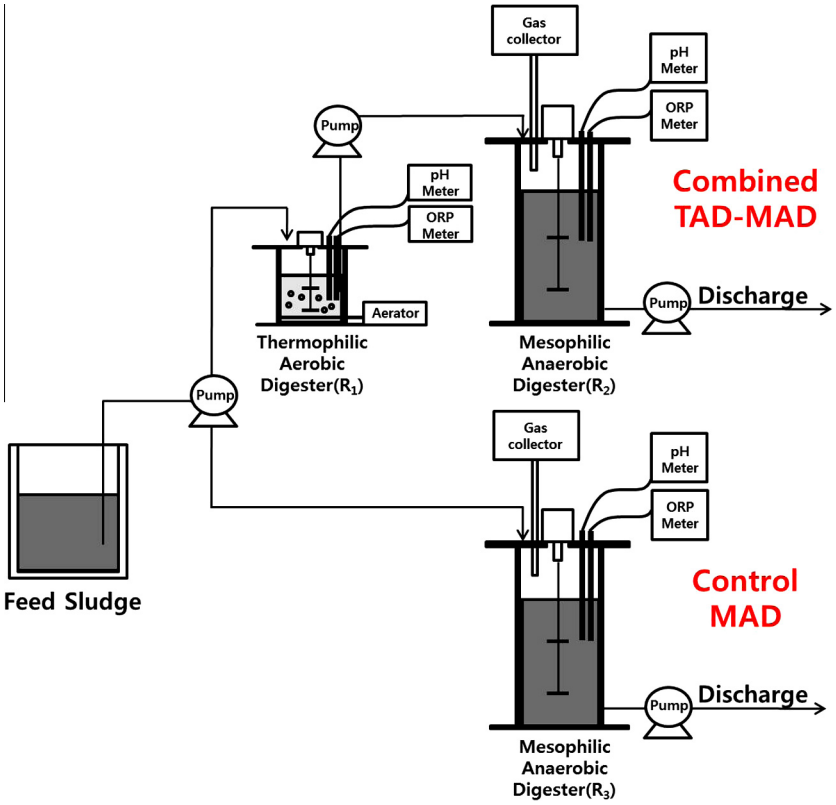


Fig. 1. Schematic diagram of combined TAD–MAD and control MAD process for sludge treatment.

Table 2
Operating conditions of combined TAD–MAD and control MAD process.

Parameter	TAD (R_1)	MAD (R_2)	Control MAD (R_3)
Reactor vol. (L)	2	7	7
Working vol. (L)	0.5	5	5
SRT (d)	4[a], 2[b], 1[c,d]	40, 39[d]	40
Airflow (L/min)	3–5	–	–
Temperature (°C)	55	35	35

[a] TAD SRT in phase I.
[b] TAD SRT in phase II.
[c] TAD SRT in phase III.
[d] TAD and MAD SRT in phase IV.

tinuously measured by pH meter (405-DPAS-SC-K85, METTLER TOLLEDO, Switzerland) and ORP meter (Pt-4805, METTLER TOLLEDO, Switzerland). Volatile fatty acids (VFAs) were quantified by high-performance liquid chromatography (HPLC, Agilent Technology 1100 series, Agilent Inc., USA) equipped with a column (Amin-ex HPX-87H, BIORAD Inc., USA), refractive index detector (RID), and diode array detector (DAD). Biogas was detected by a gas chromatograph (Model 6890N, Agilent Inc., USA) equipped with a pulsed discharge detector (PDD), and biogas volume was quantified using the water displacement method. The quantified values, including VFA, carbohydrate, and protein, were converted to g COD/L by using conversion factors (1.066: acetic acid, 1.07: carbohydrate, 1.50: protein).

2.4. Microbial community analysis

2.4.1. DNA extraction

After sampling from startup and steady state at each phase, 1 mL of sample was centrifuged twice at 13,000*g* for 5 min, and sludge pellets were washed in deionized and distilled water (DDW). The washed sample was re-suspended in 200 μL of DDW. Total DNA extraction and purification was conducted by Nucleo Spin® Soil kit (MACHEREY–NAGEL, Germany). The purified DNA was eluted with 50 μL of 1× TE buffer and stored at −25 °C for further analysis.

2.4.2. PCR amplification and DGGE analysis

The PCR–DGGE technique is useful to characterize microbial community structure within a bioreactor. Bacterial and archaeal 16S rRNA genes were amplified by PCR using universal primers (Table 3) (Shin et al., 2010). DGGE analysis was conducted with DNA samples collected at steady state in each phase (Fig. 4). PCR of bacteria samples from R_1, R_2, and R_3 successfully amplified the products. However, only R_2 and R_3 samples succeed in archaea PCR, because most of the archaea group grows under strict anaerobic conditions.

The PCR amplifications were carried out in a My Cycler™ Personal Thermal Cycler (Bio-Rad Corp., USA). The process used 50 μL of PCR reaction mixture containing 1.25 U *TaKaRa Ex Taq*™ DNA polymerase (TaKaRa Code: RR001A, TaKaRa BIO INC., Japan), 5 μL of 10× PCR buffer, 4 μL of dNTP (2.5 mM each), 2 μL of each primer (final concentration 1 μM), and 32.75 μL of PCR-grade water. The PCR step comprised: initial denaturation at 95 °C for 10 min, followed by 30 cycles of: (1) 95 °C for 5 min, (2) 55 °C for 30 s, (3) 72 °C for 30 s, and a final extension at 72 °C for 10 min. DGGE profiling was conducted by a Dcode™ Universal Mutation Detection System (Bio-Rad Corp., USA).

The PCR product was added to each well of an 8% (w/v) acrylamide gel (acrylamide: bis-acrylamide solution, 37.5:1) containing a 30–60% denaturant gradient (100% denaturant agent was defined as 7 M urea with 40% formamide). Electrophoresis was performed in 0.5× TAE buffer for 720 min at 100 V and 60 °C. Following electrophoresis, the gel was stained with 0.5× TAE buffer containing

Table 3
Characteristics of primer sets used in DGGE and real-time PCR analysis.

Target group	Primer	Sequence (5′–3′)	Amplicon size (bp)	Representative strains (culture collection)
Bacteria[a]	BAC338F	ACTCC TACGG GAGG CAG	468	*Escherichia coli* K12 (DSM 1607)
	BAC805R	GACTA CCAGG GTATC TAATC C		
Archaea[a]	ARC787F	ATTAG ATACC CSBGT AGTCC	273	*Methanomicrobium mobile* BP (DSM 1539)
	ARC1059R	GCCAT GCACC WCCTC T		
Methanobacteriales	MBT857F	CGWAG GGAAG CTGTT AAGT	343	*Methanobacterium formicicum* M.o.H. (DSM 863)
	MBT1196R	TACCG TCGTC CACTC CTT		
Methanomicrobiales	MMB282F	ATCGR TACGG GTTGT GGG	506	*Methanomicrobium mobile* BP (DSM 1539)
	MMB832R	CACCT AACGC RCATH GTTTA C		
Methanococcales	MCC495F	TAAGG GCTGG GCAAG T	337	*Methanococcus voltae* (DSM 1537)
	MCC832R	CACCT AGTYC GCARA GTTTA		
Methanosarcinales	MSL812F	GTAAA CGATR YTCGC TAGGT	354	*Methanosarcina barkeri* MS (DSM 800)
	MSL1159R	GGTCC CCACA GWGTA CC		

[a] When performing DGGE analysis, a 40-bp GC-clamp was added at the 5′ ends of BAC338F and ARC787F.

ethidium bromide. The visualized gel was then scanned using the Gel Doc™ Imaging System (Bio-Rad Corp., USA).

2.4.3. Analysis of DGGE bands and nucleotide sequence

Visual bands were excised and washed with 100 μL of PCR-grade water for 5 min. After washing, excised bands were eluted with 30 μL of TE buffer at 4 °C for 48 h. PCR amplification was conducted using 4.0 μL of mixture of extracted DNA solutions with corresponding primers without the GC-clamp (Table 3). The PCR products were purified and cloned using pGEM-T Easy Vector (Promega Corp., USA). The cloned 16S rRNA gene fragments were sequenced (SolGent Co., Ltd., Korea), and the closest phylogenetic affiliation of the sequence from DGGE gels were compared with the reference in the NCBI (National Centre for Biotechnology Information, http://www.ncbi.nlm.nih.gov) nucleotide sequence database (http://blast.ncbi.nlm.nih.gov/Blast.cgi).

2.4.4. Real-time PCR quantification

Real-time PCR analysis was used to quantify the 16S rRNA gene abundance of total bacteria, archaea group, and four major methanogenic orders (*Methanobacteriales*, *Methanococcales*, *Methanomicrobiales*, and *Methanosarcinales*). Real-time PCR amplification and fluorescence detection were performed using an Applied Biosystems 7300 Real-Time PCR system (Applied Biosystems, Forster City, USA). The analyses were performed in a 20-μL reaction mixture containing 10 μL TaKaRa SYBR® Premix Ex Taq™ (TaKaRa BIO INC., Japan), 0.4 μL of each primer (0.1 μM for bacteria, 0.2 μM for archaea and methanogens), 0.4 μL of 50× ROX reference dye, 6.8 μL of PCR-grade water, and 20 ng of template DNA. The PCR step for analysis comprised: 95 °C for 10 s, followed by 40 cycles of 95 °C for 5 s, 59 °C (56 °C for the total archaea) for 10 s, and 72 °C for 27 s (extension and fluorescence detection step). The properties of each primer set were designed in accordance with previous research (Table 3) (Shin et al., 2010). Primer concentration, and annealing temperature and time were selected via optimization procedure. All amplifications were performed in duplicate and included non-template control (NTC).

The standard curve was calculated following a procedure described in previous research. (Yu et al., 2005). The 10-fold serial-diluted 16S rRNA from the representative strains (Table 3) was used to obtain the standard curve for each primer set. The threshold cycle (C_t) value was automatically calculated by the SDS software of the 7300 Real-Time PCR system (Applied Biosystems, Forster City, USA). After amplification, melting curve (95 °C) and agarose gel electrophoresis analyses were conducted to confirm primer specificity.

3. Results and discussion

3.1. Reactor performance

3.1.1. VSS, SCOD, and methane

In this study, to verify the effects of the SRT of biological TAD pretreatment on anaerobic digestion in the combined TAD–MAD process, the TAD reactor (R_1) was operated with three different SRTs: 4-, 2- and 1-day, for a total of 276 days; MAD reactors (R_2, R_3) in phases I, II, and III were operated at 40-day SRT. In phase IV, the SRTs of TAD and MAD in the combined process were adjusted to 1-day and 39-days, respectively, to make the total control volume of the combined process the same as the control MAD (R_3).

The change in VSS concentration during digestion is shown in Fig. 2a. The VSS is a significant factor in sludge hydrolysis. Increased feed-sludge concentration and data fluctuations occur in the early stage of phase II, since the conditions of the Daegu WWTP was changed. Although some fluctuations occurred and the SRT of R_1 was decreased at each phase (Table 2), stable operation of the combined TAD–MAD (R_1–R_2) and control R_3 processes were achieved with constant VSSR efficiency. The average VSSRs of the combined TAD–MAD process and control R_3 during digestion at different phases were 57–58% and 42–45%, respectively, which indicated that TAD pretreatment enhanced VSSR. This result is supported by those of previous studies, which reported that lytic enzymes excreted by thermophilic aerobic bacteria enhanced the sludge reduction rate during anaerobic digestion (Miah et al., 2005). Furthermore, some fraction of biological flocs is only degraded under aerobic conditions (Novak et al., 2003). Hence, by applying TAD pretreatment (R_1), VSSR in the combined TAD–MAD process would be enhanced by consuming refractory organic matters which are not degradable in the MAD.

As shown in Fig. 2b, R_2 and R_3 were operated with constant SCOD concentration, in contrast to R_1. Although a large proportion of VSS was degraded, the SCOD concentration in R_1 was lower than for the feed sludge in phase I; this is explained by thermophilic microbes actively utilizing organic matter released by VSS reduction during relatively long SRT, resulting in high VSS reduction and their rapid growth (Hartman et al., 1979). However, SCOD values in R_1 increased significantly, to 8.6–12.4 g/L at phases II, III, and IV, showing that a rapid increase SCOD in phases II, III, and IV was closely related to the SRT of R_1 (Table 2). It is likely that as reducing the SRT to 1 day in R_1, TAD remained feasible with high SCOD production and stable process performance, due to the fact that the SCOD concentrations produced in terms of sludge destruction and release of intracellular materials from the cells during relatively short SRT were higher than that of consumption by thermophilic aerobic microorganisms in TAD.

The methane production rates (MPR) in R_2 and R_3 are shown in Fig. 2c. Overall, the average MPR of control R_3 was about 100 mL/L/

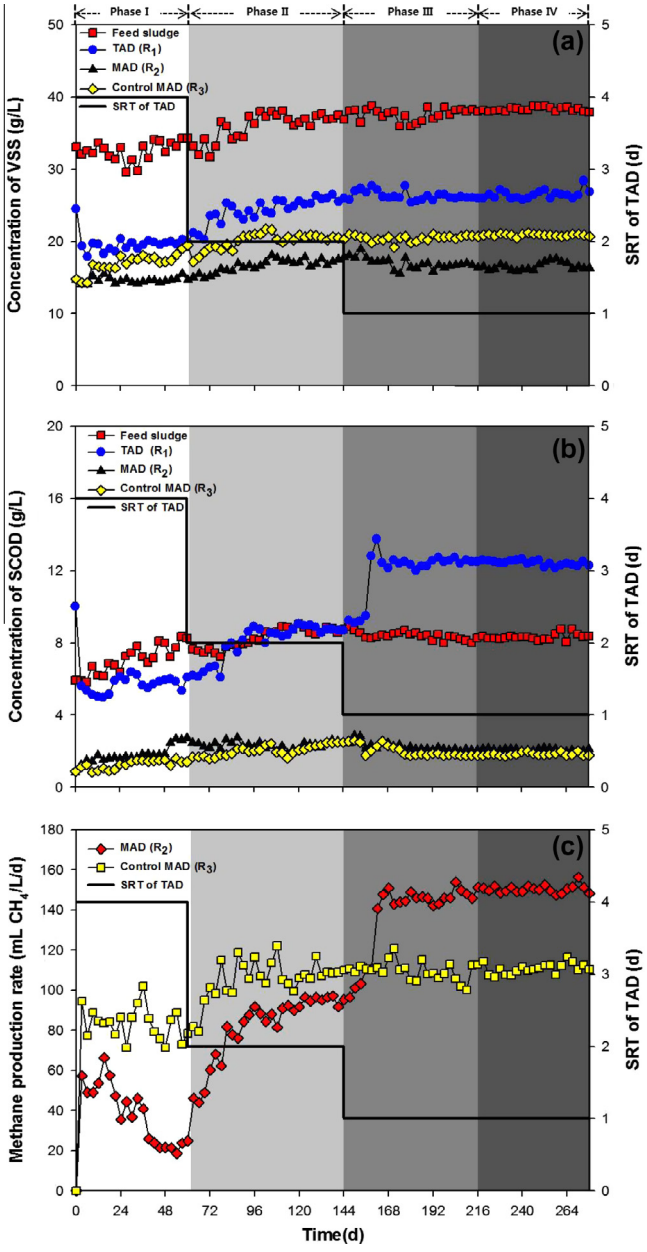


Fig. 2. Change in (a) VSS, (b) SCOD, and (c) methane production rate (MPR) during digestion.

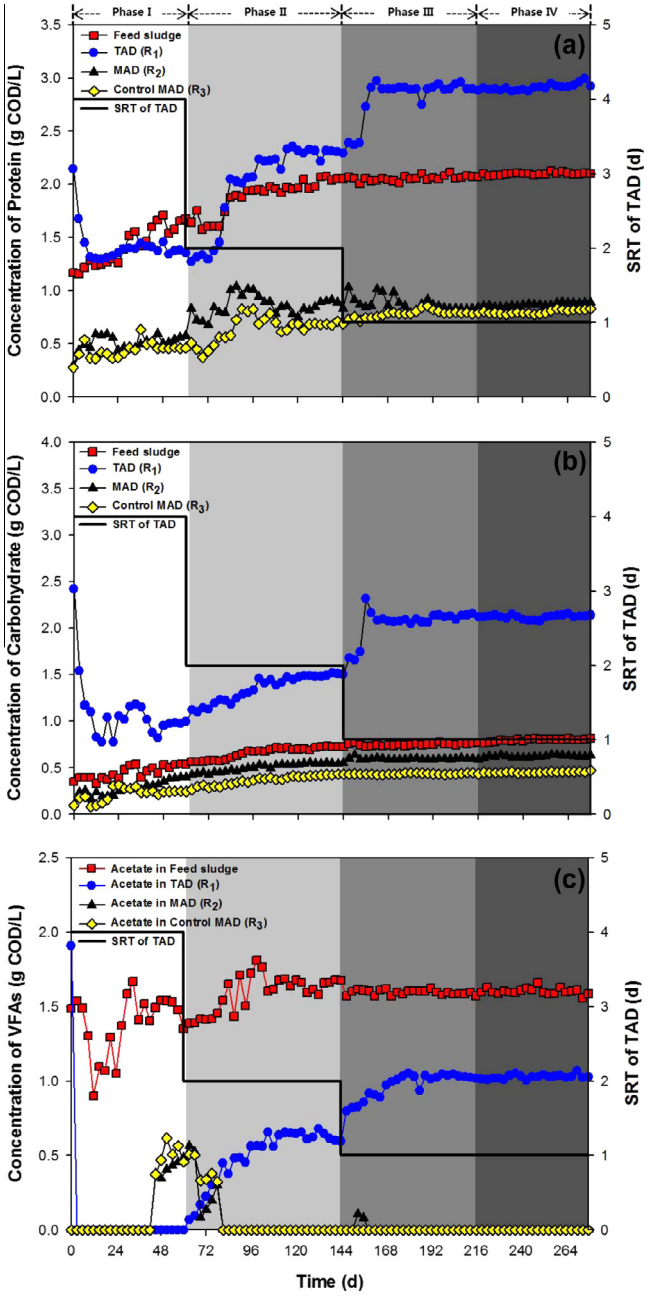


Fig. 3. Variation in (a) protein, (b) carbohydrate, and (c) VFAs during digestion.

d. In contrast, associated with successful sludge digestion, methane production in R_2 shows better performance than R_3 at phases III and IV. Under steady state condition in phase I, the average MPR of R_2 was approximately 22 mL/L/d since soluble organic matters in the TAD were mostly consumed by thermophilic microbial species due to relatively long SRT. However, after phase I, the MPR of R_2 was significantly increased with respect to the concentrations of SCOD (Fig. 2b) and VFAs (Fig. 3c) values. The average MPR of R_2 for phases II, III, and IV was about 95, 146 and 147 mL/L/d, respectively. In particular, the MPR of R_2 for phases III and IV was 46% higher than that of R_3. Meanwhile, the methane contents for phases I–IV in the combined process were 63.5%, 65.5%, 70.0%, and 71.0%, respectively, compared with 63.3% in the control R_3. Overall, the results indicate that, in R_1 during phase I, owing to relatively long SRT, the carbon consumption of the TAD process was too high to provide soluble carbon sources to the subsequent R_2; conversely, the 1-day SRT of TAD had adequate contact time between microbial enzyme and substrate to produce high levels of soluble organic matter with efficient steady-state operation, resulting in enhanced methane production. Thus, at a relatively short SRT of TAD in the combined process, the better performance of R_2 in methane production was undoubtedly ascribed to the biological TAD pretreatment.

3.1.2. Change in protein, carbohydrate, and VFA

Proteins and carbohydrates are the two of the most common forms of organic matters in extracellular polymeric substances (EPS), and comprise a large proportion of COD in the WAS. After hydrolysis of WAS, amino acids formed from proteins were transformed into a low molecular weight organic acid, ammonia, and carbon dioxide. Carbohydrates were converted to short-chain polysaccharides or reducing sugar (Qiao et al., 2011). Therefore, the

Table 4
Nitrogen species and other chemical properties of sludge during each phase (Average ± S.D.).

Parameter	Feed sludge	Control MAD (R_3)				TAD (R_1)				MAD (R_2)			
		Phase I	Phase II	Phase III	Phase IV	Phase I	Phase II	Phase III	Phase IV	Phase I	Phase II	Phase III	Phase IV
pH	6.61 ± 0.06	7.31 ± 0.04	7.34 ± 0.03	7.31 ± 0.01	7.33 ± 0.02	7.40 ± 0.14	7.66 ± 0.13	7.93 ± 0.1	8.02 ± 0.05	7.24 ± 0.07	7.32 ± 0.09	7.36 ± 0.06	7.42 ± 0.09
ORP (mV)	−235	−400	−404	−410	−408	−20–85	−20–85	−20–85	−20–84	−405	−415	−420	−411
Alkalinity (g $CaCO_3$/L)	1.72 ± 0.06	3.02 ± 0.01	3.29 ± 0.01	3.34 ± 0.02	3.42 ± 0.09	1.09 ± 0.06	1.17 ± 0.15	1.18 ± 0.1	1.15 ± 0.02	3.01 ± 0.01	3.25 ± 0.06	3.27 ± 0.04	3.24 ± 0.15
Nitrogen													
TN (g/L)	4.51 ± 0.77	2.77 ± 0.06	2.60 ± 0.1	2.80 ± 0.01	2.79 ± 0.05	2.10 ± 0.1	3.10 ± 0.2	3.11 ± 0.1	3.08 ± 0.02	1.90 ± 0.1	1.89 ± 0.06	2.04 ± 0.06	2.05 ± 0.07
STN (g/L)	0.96 ± 0.12	1.07 ± 0.03	1.29 ± 0.03	1.27 ± 0.03	1.28 ± 0.06	0.57 ± 0.04	0.65 ± 0.03	0.77 ± 0.03	0.79 ± 0.15	0.85 ± 0.01	0.84 ± 0.02	0.94 ± 0.02	0.95 ± 0.04
NH_4^+–N (g/L)	0.90 ± 0.09	1.05 ± 0.03	1.28 ± 0.03	1.21 ± 0.03	1.12 ± 0.54	0.56 ± 0.05	0.63 ± 0.02	0.72 ± 0.01	0.76 ± 0.04	0.83 ± 0.01	0.83 ± 0.02	0.88 ± 0.01	0.88 ± 0.15
NO_2^- (g/L)	–	–	–	–	–	–	–	–	–	–	–	–	–
NO_3^- (g/L)	–	–	–	–	–	–	–	–	–	–	–	–	–

measurement of proteins and carbohydrate at each step would provide a more comprehensive understanding of the effects of TAD pretreatment on R_2 performance.

Fig. 3a and b show the influence of TAD pretreatment on protein and carbohydrate. With the decrease in the SRT in R_1 from 4 days to 1 day, the concentrations of protein and carbohydrate were drastically increased to 2.88 and 2.1 g COD/L, respectively. These results were in accordance with previous results for SCOD increase (Fig. 2b). Also, the results indicated that by the biological TAD pretreatment, the amounts of protein and carbohydrate released from sludge destruction could be highly increased during the relatively short SRT in the TAD.

According to previous experimental ATAD results, VFAs can be produced in thermophilic aerobic conditions due to the low oxygen solubility and high oxygen demand (Chu et al., 1997). The production of VFAs in this study is presented in Fig. 3c. The results indicated that the only major VFA in this study was acetic acid. This finding is supported by previous studies, which reported the similar result in TAD (Mavinic et al., 2001). On the other hand, a recent study reported that acetic acid and propionic acid could be produced in one-stage batch mode ATAD (Liu et al., 2012).

The variation pattern of acetic acid is similar to SCOD (Fig. 2b), protein, and carbohydrate (Fig. 3a and b). The concentration of acetic acid rapidly increased to 0.6–1.0 g COD/L at phases II, III, and IV. Acetic acid concentration then declined sharply, and was undetected in R_2 and R_3. Based on these results, it was noted that TAD pretreatment also played an important role as a source of acetic acid when using a relatively short SRT of 1 day.

3.1.3. Variation in nitrogen species

The change in TN, as well as STN and NH_4^+–N in each phase at the steady state, are shown in Table 4. There are no significant changes of TN concentration in R_2 and R_3. However, the TN concentrations of the R_1 for phases II and III increased by 41% and 42% compared to phase I, owing to feed-sludge fluctuations. The variation in STN concentration was similar to that of NH_4^+–N. The ratio of NH_4^+–N/STN was continually maintained between 90 and 95% in all reactors, which indicated that NH_4^+–N was the major nitrogen species in STN. In general, anaerobic digestion of sludge leads to the release of ammonia (Hashimoto, 1986). Additionally, during the TAD process, degradation of protein and less temperature-tolerant cells increased NH_4^+–N concentration (Liu et al., 2010). However, the NH_4^+–N concentration at different phases was decreased by 37%, 30%, 20%, and 15% after TAD pretreatment, compared to the concentration of the feed sludge. Also, NO_2^- and NO_3^- were below detectable levels (Table 4) in R_1, since thermophilic conditions inhibited nitrification and denitrification (Willers et al., 1998). Consequently, in R_1, ammonia stripping is the only possible explanation for the decline in NH_4^+–N concentration, as there was no loss of nitrogen as nitrogen gas. In addition, the experimental results showed that thermophilic aerobic conditions and continuous aeration are favorable for ammonia stripping (Juteau et al., 2004). Therefore, the ammonium concentration of the TAD process was lower than that of the MAD process at all phases.

3.1.4. pH, alkalinity, and ORP

Variations in pH, alkalinity, and ORP are summarized in Table 4. A decline in pH within the anaerobic digester can present high potential for VFA accumulation. According to the VFAs results (Fig. 3c), there was no significant accumulation of VFAs in R_2 and R_3. Therefore, without additional pH control, the pH in the anaerobic reactors was between 7.24 and 7.36, which is within the known range for adequate anaerobic digestion. On the other hand, pH in R_1 continually increased up to around 8 at phases III and IV, owing to the stripping of ammonia and carbon dioxide. This result is also

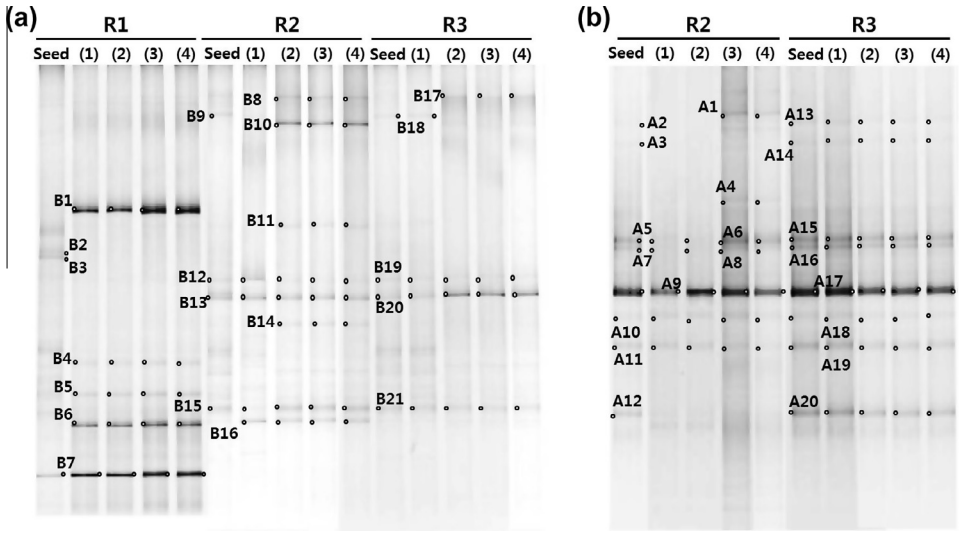


Fig. 4. DGGE profiles of 16S rRNA gene fragments from samples in R_1, R_2, and R_3: (a) bacteria profiles, (b) archaea profiles. (1)–(4) represent each phase at steady-state conditions.

supported by the variation in pH observed in a previous one-stage ATAD experiment (Liu et al., 2011).

As shown in Table 4, the alkalinity concentration in R_2 and R_3 was higher than 3 g $CaCO_3$/L during the digestion. It was found that MAD generally maintained high alkalinity concentration between 1 and 5 g $CaCO_3$/L, which was produced by the archaea group in the form of CO_2, HCO_3^-, and NH_3 (Appels et al., 2008). Also, it is likely that the higher ammonium concentrations in the MAD lead to the higher alkalinity concentrations compared to those of the influent. In contrast, R_1 maintained 1 g $CaCO_3$/L owing to alkalinity depletion by constant aeration, which caused CO_2 and NH_3 stripping.

The two MAD reactors maintained ORP values of less than −400 mV during the digestion (Table 4), indicated that R_2 and R_3 were operated under strict anaerobic conditions. On the other hand, the ORP value in R_1 shows some variations within the range −20 to 85 mV because the feed sludge was provided four times per day using the time-control relays. When the feed sludge was fed to the TAD, the ORP values declined from 85 to −20 mV, but promptly recovered to 85 mV over time. In contrast, a previous study of an ATAD system reported lower ORP values, of between −250 and −50 mV (Kelly et al., 1993). The discrepancy with the present study is probably due to differences in the experimental conditions, reactor sizes, and apparatus.

3.2. Microbial community structure and population change during digestion

3.2.1. Succession of bacterial community structure during digestion

As shown in Fig. 4a, a total of 21 visual bands were detectable from bacterial DGGEs during the digestion. All of the sequences obtained from the excised bands in R_1, and B1–B7 show strong similarity (above 90%) to GenBank reference sequences. According to phylogenetic affiliations of B1–B7, three phyla, Firmicutes, Actinobacteria, and β-Proteobacteria, were involved in R_1. As the seed sludge in R_1 underwent thermophilic aerobic conditions, two bands (B2, B3) disappeared, whereas some bands (B1, B4, B5, B6, B7) become detectable throughout the digestion. Bands B1, B6 and B7 were prominent, showing consistently high band intensity at all phases. Of these, B1 had the highest band intensity in R_1, mostly close (99%) to *Ureibacillus thermophiles* releasing lytic enzyme that is highly active in sludge degradation (Accession No. AB682456), which was reported to grow optimally at pH 7–8 and 50–60 °C (Fortina et al., 2001), compatible with R_1 conditions (Tables 2 and 4). These results were supported by previous studies, which *bacillus* species are mainly detected in the thermophilic aerobic process (Ugwuanyi et al., 2008). The B7 band was also dominant in the R_1, and showed strong similarity (92%) to Bacterium thermus (Accession No. HQ436531), a thermophilic aerobic bacterium was also related to hydrolysis and highly active to degrade organic matters in sludge during the thermophilic aerobic digestion (Liu et al., 2010). In addition, the B6 band was detected in both R_1 and R_2. B6 sequence was closely (98%) matched with *Clostridium straminisolvens* (Accession No. NR024829), which is an aerotolerant, thermophilic, anaerobic, and cellulolytic bacterium involved in hydrolysis and acidogenesis; temperature and pH for optimal growth were 50–55 °C and pH 7.5, respectively (Kato et al., 2004). Interestingly, these findings indicate that thermophilic anaerobic bacterium, *Clostridium straminisolvens* that is aerotolerant, can be viable in the thermophilic aerobic process. This may suggest that the thermophilic aerobic process in this study was not entirely aerobic condition due to oxygen dissolving and transfer limitation caused by high temperature and dense sludge concentration, even though the air was continually supplied in the TAD.

Owing to extremely different digestion conditions between the TAD and MAD reactors, the bacterial band patterns of R_2 and R_3 are in stark contrast to those of R_1 (Fig. 4a). The phylogenetic affiliations of B8–B21, the three phyla Firmicutes, Actinobacteria, and γ-Proteobacteria, were involved in the anaerobic reactors (Table 5). Overall, the seed-bacterium band patterns in R_2 were similar to R_3. Bands B12, B13, and B15 were constantly detectable during digestion. B12 was close (94%) to *Bacillus* sp. (Accession No. AJ489379), and B13 was related (95%) to *Lactobacillus amylovorus* (Accession No. NR043287), an anaerobic bacterium that can convert organic matter to lactic or acetic acid (Nakamura, 1981). According to previous studies of methanogenic bioreactors, lytic enzymes such as cellulase, protease, and other EPS can be produced by *Streptomyces* spp. and *Nocardioides* spp. (B14 and B15), which are closely related to the hydrolysis step (Levén et al., 2007). Meanwhile, various bands (B10, B11, B14, and B16) were detectable only in R_2 at phases II and III, which indicated that TAD pretreatment affects bacterial microbial structure in R_2. B8 and B10 became dominant bands at phases II and III in R_2. The sequences from B8 and B10 were mostly closely matched with uncultured *Symbiobacterium* sp. (Accession No. AB052397) and *Aeriscardovia aeriphila* (Accession No. NR042759), respectively. They can consume various forms of organic matter, such as carbohydrate and amino acid, and also

Table 5
Closet phylogenetic affiliations of sequences recovered from DGGE bands.

Band Name	Accession No.	Closest sequence	Similarity (%)	Phylogenetic group
Bacteria				
B1	AB682456	*Ureibacillus thermophilus*	99	Firmicutes
B2	JF303826	*Uncultured actinobacterium* clone HV9	95	Actinobacteria
B3	AB537980	*Coprothermobacter* sp.	98	Firmicutes
B4	DQ539621	*Petrobacter* sp.	90	β-Proteobacteria
B5	NR025725	*Petrobacter succinatimandens* sp.	100	β-Proteobacteria
B6, B16	NR024829	*Clostridium straminisolvens*	98	Firmicutes
B7	HQ436531	Bacterium thermus-lsg2	92	β-Proteobacteria
B8, B17	AB052397	Uncultured Symbiobacterium sp.	99	Firmicutes
B9, B18	FM252968	Uncultured gamma proteobacterium	95	γ-Proteobacteria
B10	NR042759	*Aeriscardovia aeriphila*	91	Actinobacteria
B11	FN435252	Uncultured bacterium	95	Unknown
B12, B19	AJ489379	*Bacillus* sp.	94	Firmicutes
B13, B20	NR043287	*Lactobacillus amylovorus*	95	Firmicutes
B14	AM889490	*Streptomyces* sp.	98	Actinobacteria
B15, B21	AB508351	*Nocardioides* sp.	95	Actinobacteria
Archaea				
A1	JF812257	*Methanosarcina* sp. MC-15	99	*Methanosarcinales*
A2, A13	JN173203	Uncultured *Methanogenium* sp.	93	*Methanomicrobiales*
A3, A14	FR836472	Uncultured *Methanomicrobiales*	98	*Methanomicrobiales*
A4	FR733698	*Methanosarcina siciliae*	99	*Methanosarcinales*
A5, A6, A15	JN004141	*Methanospirillum hungatei* strain GRAU-3	99	*Methanomicrobiales*
A7, A8, A16	FR733698	*Methanosarcina siciliae*	99	*Methanosarcinales*
A9, A17	AB679168	*Methanosaeta concilii*	99	*Methanosarcinales*
A10, A18	NR028163	*Methanolinea tarda* NOBI-1	99	*Methanomicrobiales*
A11, A19	CU917434	Uncultured *Methanosarcinales*	99	*Methanosarcinales*
A12, A20	HM630582	*Methanoculleus* sp. Annu8	98	*Methanomicrobiales*

produce low molecular weight organic matter that is beneficial to methanogen species for methane production (Simpson et al., 2004; Ueda et al., 2004). Collectively, the bacteria DGGE results indicated that the dominant thermophilic microbial species in R_1 which were highly efficient to degrade the organic matters including cells in the sludge and improve the rate-controlling step of hydrolysis were successfully enriched, and consequently the various bacteria species enhancing the subsequent acidogenesis and acetogenesis in R_2 could be enriched by improving hydrolysis of complex organics with the help of biological TAD pretreatment.

3.2.2. Succession of archaea community structure during digestion

The analysis of archaea community structures is important because most of the archaea group is involved in methane production. Archaea DGGE in R_2 and R_3 is shown in Fig. 4b. During digestion, a total of 20 visual bands were detectable and excised. With regard to the phylogenetic affiliations of A1–A20, the two orders, *Methanosarcinales* and *Methanomicrobiales*, were involved in R_2 and R_3. The A9 band *Methanosaeta concilii*, a strict aceticlastic methanogen, showed highest band intensity in both anaerobic reactors at all phases.

Overall, the band pattern of R_3 was quite stable, since the operating conditions and feed sludge concentration were not changed. On the other hand, some bands in R_2 were affected by R_1 operating conditions. As feed sludge in R_2 underwent anaerobic digestion, A2, A3, and A12 disappeared. In addition, the band intensities of A5 and A7 were significantly decreased in phases I and II, but then recovered in phases III and IV. The high archaeal diversity observed in R_2, resulted in relatively low concentration of soluble organic matter (Figs. 2 and 3) and high alkalinity (Table 4).

Acetic acid is one of the most important compounds in AD, because it can be converted to methane gas by aceticlastic methanogens, in this case *Methanosarcinales*. As the SRT of R_1 was decreased from 4 to 1 day, the soluble organic matter, including acetic acid, increased dramatically (Figs. 2 and 3). These results precisely encountered with MPR increase at phases II, III, and IV in R_2 (Fig. 2c). Also, bands A5, A7, and A9 intensified, and group A1 and A4, which are related to acetate-utilizing methanogens, appear in R_2. Overall, the archaea DGGE results indicated that the aceticlastic methanogens, especially *Methanosaeta concilii*, showed significant growth, and several *Methanosarcinales* and *Methanomicrobiales* were enriched in R_2.

3.2.3. Quantitative bacteria and archaea population dynamics

To complement the results of DGGE analysis and obtain comprehensive insight into the microbial population dynamics, real-time PCR analysis was conducted on DNA samples from the digester. Changes in the 16S rRNA gene-concentration target group, including two domains and four major methanogenic orders, are displayed in Fig. 5. Total bacteria 16S rRNA gene concentration was quantified as 6.78×10^{10} to 1.05×10^{11} copies/mL in (R_1), 2.61×10^{10} to 4.01×10^{10} in (R_2), and 2.74×10^{10} to 3.26×10^{10} in (R_3), respectively. Although, R_1 has a relatively short SRT, it maintained high bacteria population without wash-out during the digestion. Also, as shown in Fig. 5a, the 16S rRNA gene concentration of total bacteria at phase III and IV in R_2 was slightly increased since the increased soluble organic products by the biological TAD pretreatment might be easily further utilized for the growth of bacteria species in R_2.

Variation in the 16S rRNA gene concentration of total archaea and four major methanogenic orders are shown in Fig. 5b and c for R_2 and R_3, respectively. The sum of the 16S rRNA gene concentrations detected by four methanogenic order primer sets should be equal to the result of the total archaea primer set. The summed 16S rRNA gene concentrations for the four orders were 97.4 ± 1.5% and 97.9 ± 1.4% of the total archaea primer set in R_2 and R_3, respectively. This result indicated that all primer sets used in this study show high specificity to the targeted groups. The two hydrogenotrophic orders (*Methanobacteriales* and *Methanomicrobiales*) and a single order of aceticlastic methanogen (*Methanosarcinales*) were detected in both anaerobic reactors. *Methanosarcinales* was the most abundant methanogenic order in R_2 and R_3 during the overall digestion, accounting for up to 92% (at 102 days) and 82% (at 180 days) of the total archaea of 16S rRNA gene concentration in R_2 and R_3, respectively. This result may be attributed to the long SRT (40 days) in R_2 and R_3, since long SRT (of more than 20 days)

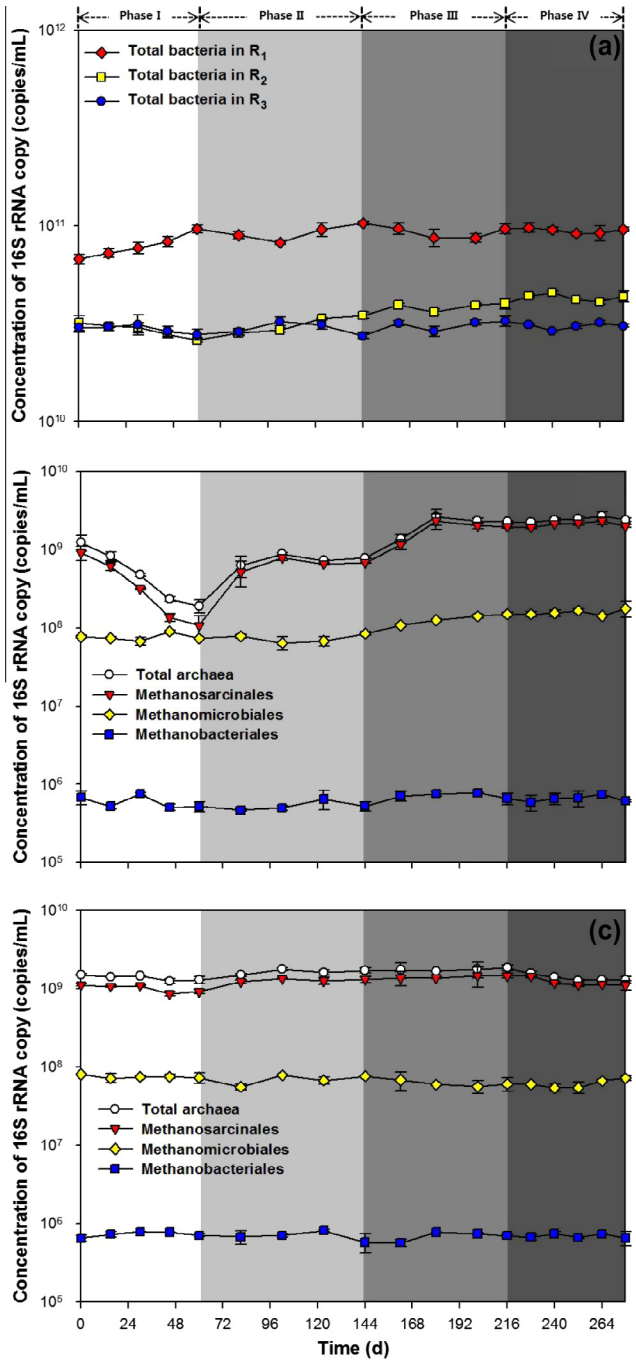


Fig. 5. Quantitative change in total bacteria and archaea groups including four major methanogens: (a) total bacteria, (b) archaea group in R_2, (c) archaea group in R_3.

is more favorable to the growth of aceticlastic methanogens than hydrogenotrophic methanogens in MAD (Bonin and Boone, 2006).

The variation in 16S rRNA gene concentration of *Methanosarcinales* and *Methanomicrobiales* in R_2 corresponded to DGGE band patterns and intensity. In R_2, the 16S rRNA gene concentration of *Methanosarcinales* decreased by 89% during phase I, from 9.18×10^8 to 1.07×10^8 copies/mL; it then increased continuously, up to 2.03×10^9 copies/mL (19-fold increase from the end of phase I) during phases III and IV. Hydrogenotrophic methanogens (*Methanomicrobiales* and *Methanobacteriales*) maintained relatively low populations (<10% of total archaea) in R_2 without washing out overall digestion and the 16S rRNA gene concentrations of *Methanomicrobiales* and *Methanobacteriales* were relatively constant, at around 10^8 and 10^5 copies/mL, respectively. Meanwhile, there was no significant change in the 16S rRNA gene concentration of all targeted archaea and the methanogenic group in control R_3.

Overall, qualitative DGGE analysis corresponded with real-time PCR results during digestion. However, some discrepancy was found between DGGE and real-time PCR analysis. No DGGE band was detected for *Methanobacteriales*, whereas it was successfully quantified by real-time PCR. The low population (<0.06% of the total archaea) might have been responsible for the insufficient band intensity in DGGE (Shin et al., 2010). The DGGE and real-time PCR results show that the contribution of thermophilic aerobic bacteria in R_1 and mesophilic bacteria in R_2 successfully enriched *Methanosarcinales*, the major group in the combined process, leading to enhanced aceticlastic methanogenesis. In particular, the importance of bacterial species in R_1 was observed for the enhancement of methane production in R_2. Also, although the hydrogenotrophic methanogens showed small populations, the experimental results indicated that the maintained abundance of hydrogenotrophic methanogens could also contribute to the enhanced production of methane in the combined process. Consequently, it is speculated that the enhanced COD removal and methane production in the combined TAD–MAD process might be attributed to the increased synergism among aceticlastic methanogens, hydrogenotrophic methanogens and bacteria species by improving the hydrolysis of rate-limiting step in sludge with the help of the biological TAD pretreatment. In this study, quantitative analysis was only performed for methanogen species. Therefore, further quantitative analysis of bacterial species will be conducted to confirm the individual contribution to the degradation of organic matter in the combined TAD–MAD process.

4. Conclusions

The effect of TAD pretreatment on MAD performance and microbial population dynamics were investigated using a lab-scale combined process. As the SRT in the TAD was shortened from 4 to 1 day, the VSSR and MPR in the MAD were much more effective than in the control MAD. Based on the DGGE results, two major bacteria species in the TAD were dominated, producing high levels of soluble organic matter and consequently enhancing interrelated microbial activities in R_2. Also, the real-time PCR results demonstrated that with the TAD pretreatment, *Methanosarcinales* increased significantly in the MAD, leading to highly increased methane production.

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